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1.

Use a decision structure to write an appropriate statement for each of the following:

a) Display Great job! when grade is 90 or higher.

A: if (grade >= 90) System.out.println("Great job!");

b) Display Error when number is less than 20 or greater than 50.

A: if (number < 20 || number > 50) System.out.println("Error");

c) Add 2 to the value of y when y is less than 100.

A: if (y < 100) y += 2;

2.

Assume num1 and num2 contain integer values. Write an if-else if statement that displays one of the following messages as appropriate:

```
First number is larger.
Second number is larger.
Numbers are equal.
```

A:

```
if (num1 > num2) {
    System.out.println("First number is larger.");
} else if (num2 > num1) {
    System.out.println("Second number is larger.");
} else {
    System.out.println("Numbers are equal.");
}
```

3.

a) Which is the appropriate word, odd or even for the blanks below?

```
if (num % 2 == 0) { System.out.println("___ number");
} else {
System.out.println("___ number")
}
```

A: blank 1 = Even, blank 2 = odd

b) Rewrite the if-else as a switch statement.

A:

```
switch (num % 2) {
        case 0:
                System.out.println("even number");
                break;
        default:
                System.out.println("odd number");
                break;
        }
```

4.

The nextInt() method in the Random class generates a random integer between 0 and a specified maximum value. Write a formula that includes the nextInt() method for each of the following situations:

a) Generate a random integer between 1 and 50.

A: rand.nextInt(50) + 1;

b) Generate a random integer between 20 and 100.

A:rand.nextInt(81) + 20;

c) Generate a random double between 10 and 20, inclusive.

A:rand.nextDouble() * 10.0 + 10.0;

5.

Identify the logic errors in the statements below, which should display a single appropriate message for any value of age:

```
if (age < 18) { System.out.println("child");
} else if (age > 18 && age < 65) {
System.out.println("adult");
} else if (age > 65) {
System.out.println("senior");
}
```

A: else if statement is missing >=, <=

6. Given the following assignments, determine if each of the following expressions evaluates to true or false:

size = 100 weight = 50 value = 75

a) size > 50 && weight == 50 **A: true**

b) value < 100 && !(weight == 50) **A: false**

c) size >= 100 || value >= 100 **A: true**

d) weight < 50 || size > 50 **A: true**

e) !(value < 75) **A: true**

f) !(size > 100 && weight >50 && value > 75) **A: True**

g) (value < 125 || weight < 76) && size ==100 **A: true**

8.

Determine if each of the following are true or false. If false, explain why.

a) The condition of an if statement must be a Boolean expression.

A: True

b) A roundoff error can occur when comparing two integers.

A: False, roundoff affect only double/float

c) A nested if statement and an if-else if statement are the same.

A: False

d) The expression in a switch statement must evaluate to a double.

A: false

e) Numbers generated by a computer program are actually pseudorandom.

A: true

f) Specifying a seed value results in a different sequence of “random” numbers each time the program is run.

A: false

g) A compound Boolean expression can contain more than two Boolean expressions.

A: true

h) In a logical And expression, both operands must be true for the expression to evaluate to true.

A: true

i) In logical expressions, && is evaluated before !.

A: false, ! evaluated before &&

j) The pow() method in the Math class is used for exponentiation.

A: true

k) The statement x = abs(-3); will return the value 3.

A: false, need to use Math.abs(-3);

l) A diamond shaped object represents a decision in a flowchart

A: true